

```

1  /"=====
2  BOOK_FIXED_DEPOSIT
3  ====="/
4  CREATE OR ALTER PROCEDURE BookFixedDeposit
5  (
6  @AccountID INT,
7  @DepositAmount DECIMAL(18,2),
8  @InterestRate DECIMAL(5,2),
9  @TenureMonths INT,
10 @NomineeID INT = NULL,
11 @Channel VARCHAR(20) = 'Branch'
12 )
13 AS
14 BEGIN
15     SET NOCOUNT ON;
16
17     BEGIN TRY
18         BEGIN TRANSACTION;
19
20         -- 1. Validate deposit amount
21         IF @DepositAmount <= 0
22         BEGIN
23             THROW 50001, 'FD deposit amount must be greater than zero.', 1;
24         END;
25
26         -- 2. Validate interest rate
27         IF @InterestRate <= 0
28         BEGIN
29             THROW 50002, 'Interest rate must be greater than zero.', 1;
30         END;
31
32         -- 3. Validate tenure
33         IF @TenureMonths <= 0
34         BEGIN
35             THROW 50003, 'Tenure must be greater than zero months.', 1;
36         END;
37
38         -- 4. Validate channel
39         IF @Channel NOT IN
40         (
41             'Branch',
42             'Internet Banking',
43             'Mobile Banking'
44         )
45         BEGIN
46             THROW 50004, 'Invalid channel for Fixed Deposit booking.', 1;
47         END;
48
49         -- 5. Validate account
50         IF NOT EXISTS
51         (
52             SELECT 1
53             FROM Accounts
54             WHERE AccountID = @AccountID
55                   AND AccountStatus = 'Active'
56         )
57         BEGIN
58             THROW 50005, 'Account does not exist or is not active.', 1;
59         END;
60
61         -- 6. Get account balance with locking
62         DECLARE
63             @OpeningBalance DECIMAL(18,2),
64             @AvailableBalance DECIMAL(18,2);
65
66         SELECT
67             @OpeningBalance = CurrentBalance,
68             @AvailableBalance = AvailableBalance
69         FROM Accounts WITH (UPDLOCK, HOLDLOCK)
70         WHERE AccountID = @AccountID;
71
72         -- 7. FD must be funded using available own balance
73         -- Overdraft is NOT considered for FD booking
74
75         IF @AvailableBalance < @DepositAmount
76         BEGIN
77             THROW 50006, 'Insufficient available balance to book this Fixed Deposit.', 1;
78         END;
79
80         -- 8. Validate nominee ownership
81         IF @NomineeID IS NOT NULL
82         BEGIN
83             IF NOT EXISTS
84             (
85                 SELECT 1
86                 FROM Nominees N
87                 INNER JOIN Accounts A
88                     ON A.CustomerID = N.CustomerID
89                 WHERE N.NomineeID = @NomineeID
90                       AND A.AccountID = @AccountID
91             )
92             BEGIN
93                 THROW 50007, 'Nominee does not belong to this account holder.', 1;
94             END;
95         END;
96
97         -- 9. Generate FD dates
98         DECLARE @StartDate DATE;
99         SET @StartDate = CAST(SYSDATETIME() AS DATE);
100
101         DECLARE @MaturityDate DATE;
102         SET @MaturityDate = DATEADD(MONTH, @TenureMonths, @StartDate);
103
104         -- 10. Calculate maturity amount using function
105         DECLARE @MaturityAmount DECIMAL(18,2);
106         SET @MaturityAmount =
107             dbo.Calculate_FD_Maturity
108             (
109                 @DepositAmount,
110                 @InterestRate,
111                 @TenureMonths
112             );
113
114         -- 11. Generate FD number
115         DECLARE @FDNumber VARCHAR(20);
116
117         SET @FDNumber = 'FD-' + CONVERT(VARCHAR(8), CAST(SYSDATETIME() AS DATE), 112)
118             + '-'
119             + RIGHT('000000' + CAST(NEXT VALUE FOR TransactionReferenceSeq
120                                     AS VARCHAR(6)), 6);
121
122         -- 12. Debit account
123         UPDARE Accounts
124         SET
125             CurrentBalance = CurrentBalance - @DepositAmount,
126             AvailableBalance = AvailableBalance - @DepositAmount,
127             LastTransactionDate = SYSDATETIME(),
128             ModifiedDate = SYSDATETIME()
129         WHERE AccountID = @AccountID;
130
131         -- 13. Insert Fixed Deposit
132         INSERT INTO FixedDeposits
133         (
134             AccountID,
135             FDNumber,
136             DepositAmount,
137             InterestRate,
138             TenureMonths,
139             StartDate,
140             MaturityDate,
141             MaturityAmount,
142             NomineeID,
143             Status
144         )
145         VALUES
146         (
147             @AccountID,
148             @FDNumber,
149             @DepositAmount,
150             @InterestRate,
151             @TenureMonths,
152             @StartDate,
153             @MaturityDate,
154             @MaturityAmount,
155             @NomineeID,
156             'Active'
157         );
158
159         -- 14. Insert financial transaction
160         INSERT INTO Transactions
161         (
162             TransactionReferenceNumber,
163             TransferReferenceNumber,
164             AccountID,
165             TransactionType,
166             TransactionCategory,
167             PaymentChannel,
168             Amount,
169             Charges,
170             TaxAmount,
171             OpeningBalance,
172             ClosingBalance,
173             CounterpartyAccountNumber,
174             CounterpartyName,
175             CounterpartyBank,
176             TransactionStatus,
177             Remarks,
178             InitiatedBy,
179             ApprovedBy,
180             TransactionDate,
181             CreatedDate
182         )
183         VALUES
184         (
185             'FDB-'
186             + CONVERT(VARCHAR(8), CAST(SYSDATETIME() AS DATE), 112)
187             + '-'
188             + RIGHT('000000' + CAST(NEXT VALUE FOR TransactionReferenceSeq AS VARCHAR(6)), 6),
189             NULL,
190             @AccountID,
191             'Debit',
192             'FD Booking',
193             @Channel,
194             @DepositAmount,
195             0.00,
196             0.00,
197             @OpeningBalance,
198             @OpeningBalance - @DepositAmount,
199             NULL,
200             NULL,
201             'Success',
202             'Fixed Deposit ' + @FDNumber + ' booked via ' + @Channel,
203             NULL,
204             NULL,
205             SYSDATETIME(),
206             SYSDATETIME()
207         );
208
209         -- 15. Commit complete FD booking
210         COMMIT TRANSACTION;
211
212         -- 16. Return result
213         SELECT
214             @FDNumber AS FDNumber,
215             @AccountID AS AccountID,
216             @DepositAmount AS DepositAmount,
217             @InterestRate AS InterestRate,
218             @TenureMonths AS TenureMonths,
219             @StartDate AS StartDate,
220             @MaturityDate AS MaturityDate,
221             @MaturityAmount AS MaturityAmount,
222             'Fixed Deposit booked successfully.' AS Message;
223     END TRY
224     GO

```

SP_BookFixedDeposits.sql - not connected* SQLQuery18.sql - (SATYAM\ABHI (52)) * SQLQuery16.sql - not connected* SP_CashDeposit.sql - not connected TRG_FixedDeposits.sql - not connected Test_queries.sql - not connected

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-- Example : Book Fixed Deposit through Mobile Banking
EXEC BookFixedDeposit
    @AccountID = 25, @DepositAmount = 20000, @InterestRate = 6.75,
    @TenureMonths = 12, @NomineeID = NULL, @Channel = 'Mobile Banking';
-- Example : Book Fixed Deposit through Branch
EXEC BookFixedDeposit
    @AccountID = 47, @DepositAmount = 39000, @InterestRate = 5.20,
    @TenureMonths = 14, @NomineeID = 25, @Channel = 'Branch';
-- Example : View updated account balances
select * from Accounts
where AccountID IN (25,47)
-- Example : View FD booking transactions
select * from Transactions
where AccountID IN (25,47)
-- Example : View newly created Fixed Deposits
select * from FixedDeposits
where AccountID IN (25,47)
-- Example : View audit records
select * from Auditing
```

AccountID	AccountNumber	CustomerID	BranchID	AccountTypeID	OpeningDate	CurrentBalance	AvailableBalance	LoanAmount	OverdraftLimit	CurrencyCode	AccountStatus	FreezeReason	ClosedDate	ClosedByEmployeeID	LastTransactionDate	CreatedDate	ModifiedDate
25	BSGB007000025	24	7	3	2017-03-28	200779.87	200779.87	0.00	100000.00	INR	Active	NULL	NULL	NULL	2026-08-14 21:39:57.3183683	2026-08-09 17:07:30.1766834	2026-08-14
47	BSGB009000047	42	6	7	2016-05-27	619014.90	619014.90	0.00	100000.00	INR	Active	NULL	NULL	NULL	2026-06-11 06:00:00.0000000	2026-08-09 17:07:30.1766834	2026-08-14

TransactionID	TransactionReferenceNumber	TransferReferenceNumber	AccountID	TransactionType	TransactionCategory	PaymentChannel	Amount	Charges	TaxAmount	OpeningBalance	ClosingBalance	CounterpartyAccountNumber	CounterpartyName	CounterpartyBank	TransactionStatus	Remark
38	DEP-20260814-000038	NULL	25	Credit	Cash Deposit	Branch	5000.00	0.00	0.00	195779.87	200779.87	NULL	NULL	NULL	Success	Deposit

FDID	AccountID	FDNumber	DepositAmount	InterestRate	TenureMonths	StartDate	MaturityDate	MaturityAmount	NomineeID	Status
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AuditID	TableName	RecordID	OperationType	OldValue	NewValue	ChangedBy	ChangedDate	IPAddress	EmployeeID
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Before
FD Booking

Query executed successfully. SATYAM\SQLEXPRESS (16.0 RTM) SATYAM\ABHI (52) BankingSystem 00:00:00 3 rows

FDNumber	AccountID	DepositAmount	InterestRate	TenureMonths	StartDate	MaturityDate	MaturityAmount	Message
FD-20260815-000042	25	20000.00	6.75	12	2026-08-15	2027-08-15	21350.00	Fixed Deposit booked successfully.
FD-20260815-000044	47	39000.00	5.20	14	2026-08-15	2027-10-15	41366.00	Fixed Deposit booked successfully.

AccountID	AccountNumber	CustomerID	BranchID	AccountTypeID	OpeningDate	CurrentBalance	AvailableBalance	LoanAmount	OverdraftLimit	CurrencyCode	AccountStatus	FreezeReason	ClosedDate	ClosedByEmployeeID	LastTransactionDate	CreatedDate	ModifiedDate
25	BSGB007000025	24	7	3	2017-03-28	180779.87	180779.87	0.00	0.00	INR	Active	NULL	NULL	NULL	2026-08-15 12:43:33.9458762	2026-08-09 17:07:30.1766834	2026-08-15
47	BSGB009000047	42	6	7	2016-05-27	580014.90	580014.90	0.00	100000.00	INR	Active	NULL	NULL	NULL	2026-08-15 12:43:33.9458762	2026-08-09 17:07:30.1766834	2026-08-15

TransactionID	TransactionReferenceNumber	TransferReferenceNumber	AccountID	TransactionType	TransactionCategory	PaymentChannel	Amount	Charges	TaxAmount	OpeningBalance	ClosingBalance	CounterpartyAccountNumber	CounterpartyName	CounterpartyBank	TransactionStatus	Remark
38	DEP-20260814-000038	NULL	25	Credit	Cash Deposit	Branch	5000.00	0.00	0.00	195779.87	200779.87	NULL	NULL	NULL	Success	Deposit
42	FDB-20260815-000043	NULL	25	Debit	FD Booking	Mobile Banking	20000.00	0.00	0.00	200779.87	180779.87	NULL	NULL	NULL	Success	Fixed C
43	FDB-20260815-000045	NULL	47	Debit	FD Booking	Branch	39000.00	0.00	0.00	619014.90	580014.90	NULL	NULL	NULL	Success	Fixed C

FDID	AccountID	FDNumber	DepositAmount	InterestRate	TenureMonths	StartDate	MaturityDate	MaturityAmount	NomineeID	Status
63	25	FD-20260815-000042	20000.00	6.75	12	2026-08-15	2027-08-15	21350.00	NULL	Active
64	47	FD-20260815-000044	39000.00	5.20	14	2026-08-15	2027-10-15	41366.00	25	Active

After
FD Booking

AuditID	TableName	RecordID	OperationType	OldValue	NewValue	ChangedBy	ChangedDate	IPAddress	EmployeeID
38	FixedDeposits	63	INSERT	NULL	FDNumber=FD-20260815-000042; AccountID=25; Amount...	SATYAM\ABHI	2026-08-15 12:43:33.9501385	NULL	NULL
37	FixedDeposits	64	INSERT	NULL	FDNumber=FD-20260815-000044; AccountID=47; Amount...	SATYAM\ABHI	2026-08-15 12:43:33.9458762	NULL	NULL